BANSAL INSTITUTE OF SCIENCE AND TECHNOLOGY BHOPAL

Department of Computer Science and Engineering

Machine Learning CS 601

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Machine Learning

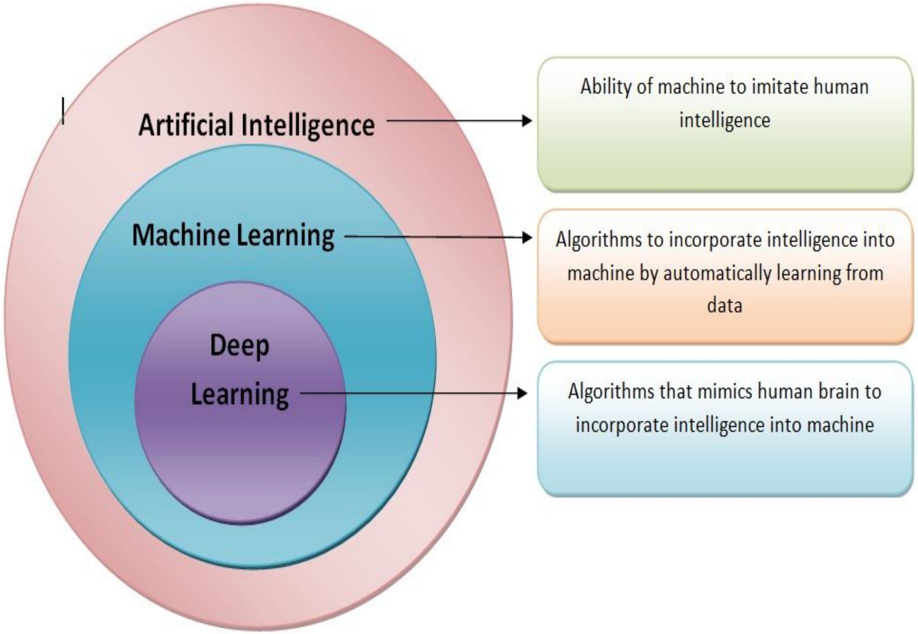
UNIT I

What is AI?

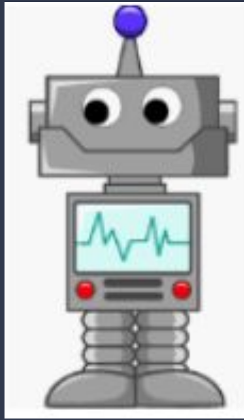
Artificial intelligence (AI) is a field that enables computers to process data in a way that mimics the human brain.

It was coined by John McCarthy in 1956

AI vs ML vs DL



What is Machine learning?



?

"Learning is any process by which a system improves performance from experience." - **Herbert Simon**

Definition by **Tom Mitchell (1998)**:
Machine Learning is the study of algorithms that

- improve their performance
- at some task T
- with experience E.

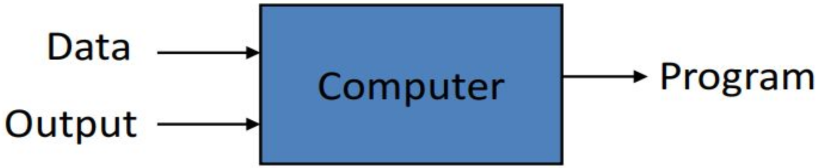
A well-defined learning task is given by.

"It was coined by Arthur Samuel in 1959."

Machine learning

Traditional Programming

Machine Learning



Machine Learning Continue

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Dataset

+

Then, devise algorithms and execute programs on the data With respect to the underlying target tasks

Then, using the programs, Identify required rules

Extract required patterns

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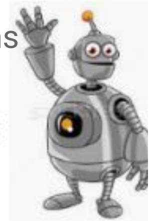
Identify relations

=



So that machine can derive inferences from the data

Scope and Limitation of Machine Learning

Scope:

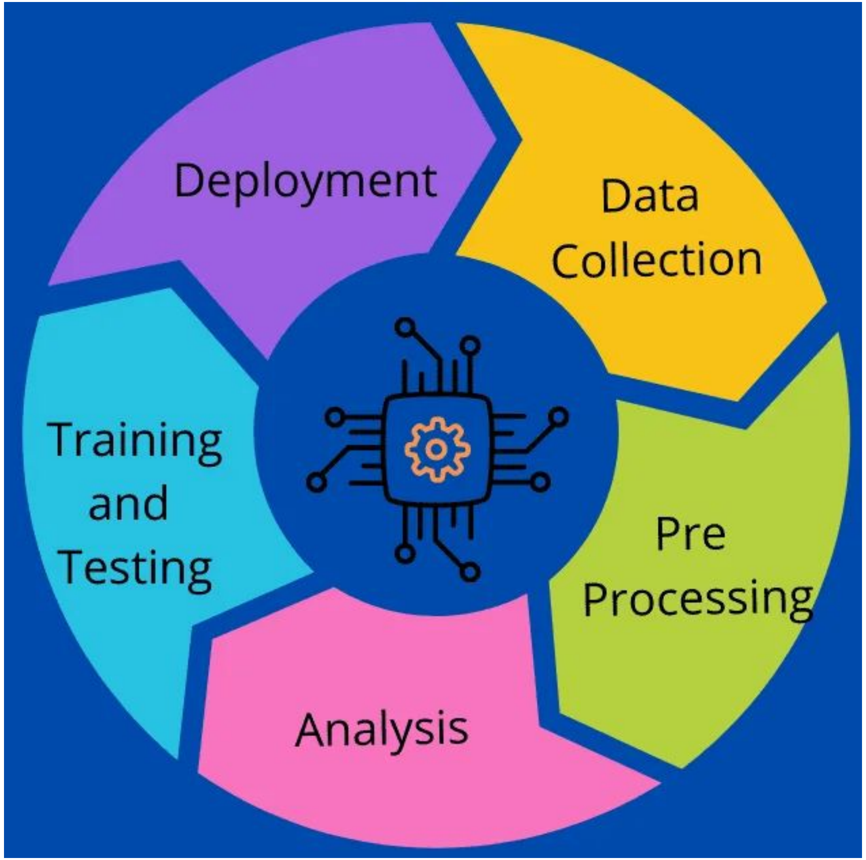
- Education
- Agriculture
- Healthcare
- Banking
- Transport
- Cyber Security
- Digital Marketing
- Stock Market
- Search Engine etc.

Limitation:

- Limited Data Availability
- Lack of good data
  - Preprocessing required
  - Heterogeneity
  - No variation
- Deal in multidimensional data
- Lack of Transparency and Interpretability
- Overfitting and Underfitting
- Code of Ethics
- Computational Resources

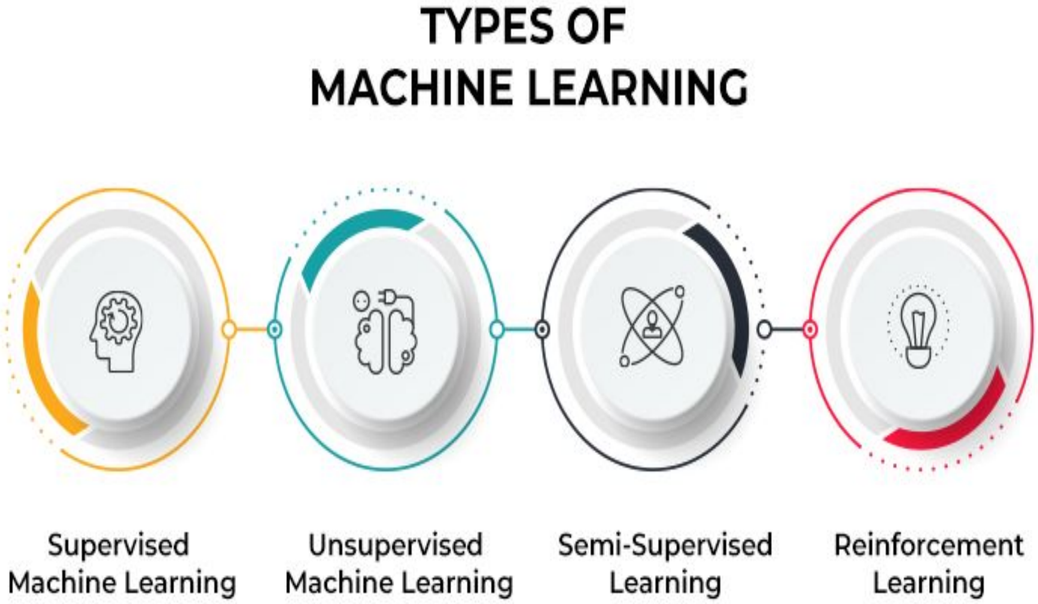
Machine Learning Life Cycle

- Data Collection
  - Identify various data sources
  - Collect data
- Preprocessing
  - Data exploration
  - Data Cleaning
    - Missing Values
    - Duplicate data
    - Invalid data
    - Noise
- Analysis
  - Selection of analytical techniques
- Train the model
  - Building models
  - Review the result
- Test the model
  - prediction
- Deployment

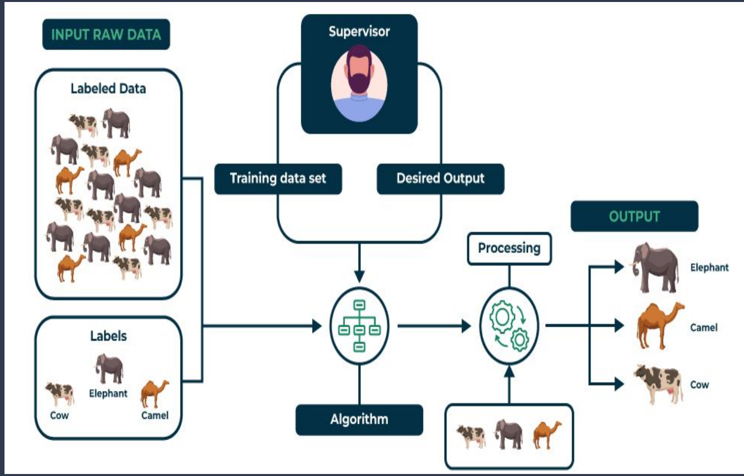


Types of Machine Learning

- Supervised Machine Learning
- Unsupervised Machine Learning
- Semi-Supervised Machine Learning
- Reinforcement Learning



Supervised Learning



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Supervised learning is defined as when a model gets trained on a "labeled dataset.". Labeled datasets have both input and output parameters. It has both training and validation datasets labeled.

There are two main categories of supervised learning that are mentioned below:

Classification

Classification deals with predicting categorical target variables, which represent discrete classes or labels.

Here are some classification **algorithms:**

- Logistic Regression
- Support Vector Machine
- Random Forest
- Decision Tree
- K-Nearest Neighbors (KNN)
- Naive Bayes

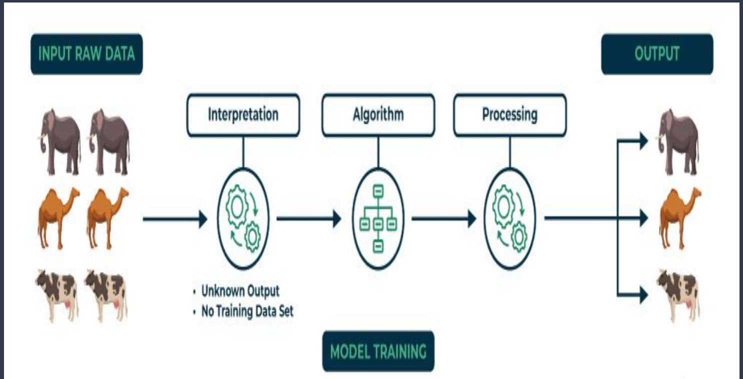
Regression

Regression, on the other hand, deals with predicting continuous target variables, which represent numerical values.

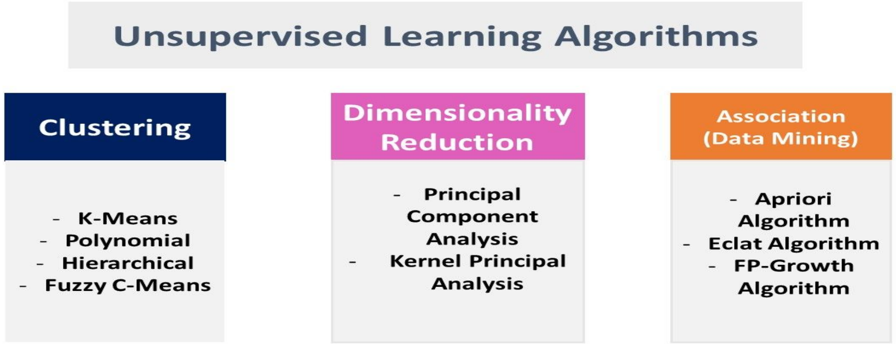
Here are some regression **algorithms:**

- Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression
- Decision tree
- Random Forest

Unsupervised Learning



- **Unsupervised Learning** Unsupervised learning is a type of machine learning technique in which an algorithm discovers patterns and relationships using unlabeled data.
- Unlike supervised learning, unsupervised learning doesn't involve providing the algorithm with labeled target outputs.
- The primary goal of Unsupervised learning is often to discover hidden patterns, similarities, or clusters within the data, which can then be used for various purposes, such as data exploration, visualization, dimensionality reduction, and more.



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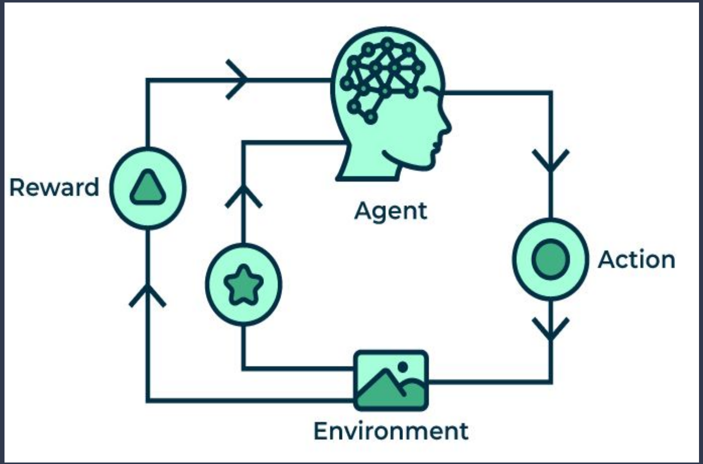
Reinforcement Learning



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- **Reinforcement Learning** trains a machine to take suitable actions and maximize its rewards in a particular situation.
- It uses an agent and an environment to produce actions and rewards.
- The agent has a start and an end state. But, there might be different paths for reaching the end state, like a maze.
- In this learning technique, there is no predefined target variable.

Algorithms

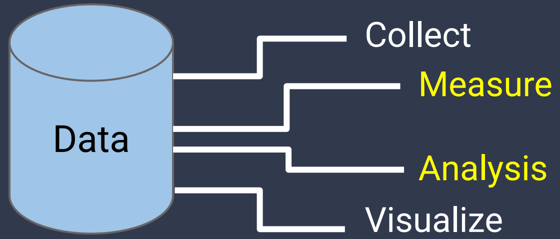
Some of the important reinforcement learning algorithms are:

1. Q-learning
2. Sarsa
3. Monte Carlo
4. Deep Q network

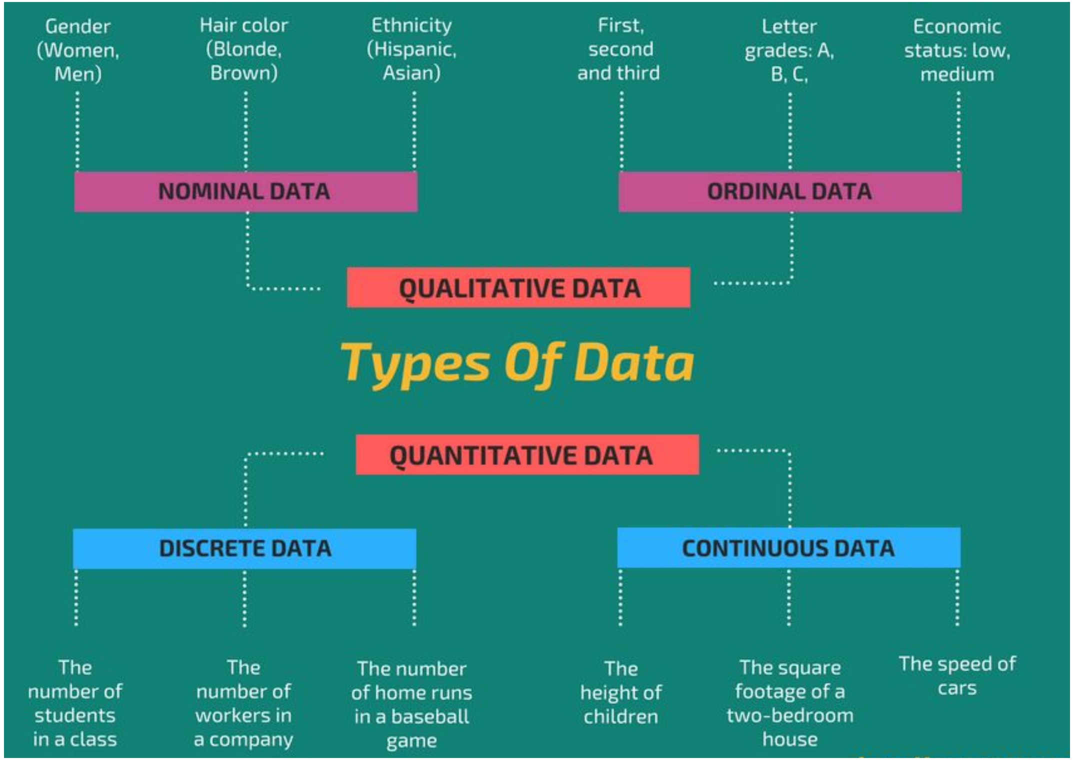
Statistics & Probabilities

What is Data ?

Data is defined as the collection of numbers, characters, images, and others that can arranged in some manner to form meaningful information. In statistics, the data is mainly the collection of numbers that is first studied then analyzed and presented in some way that we can get some meaningful insight from that data.



Types of Data



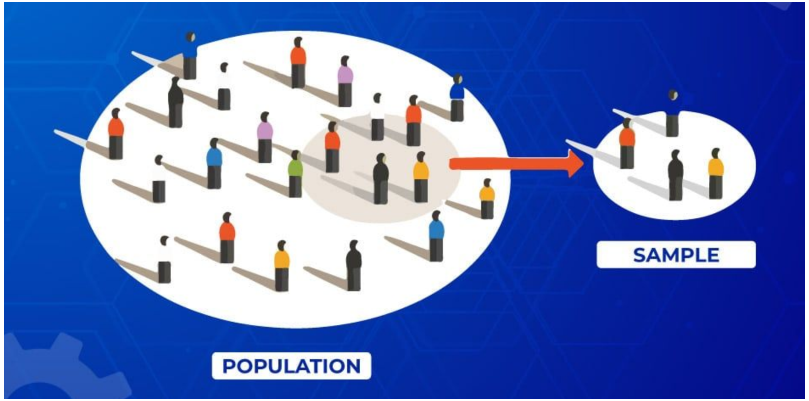
Statistics

Statistics is the area of applied mathematics concerned with the data collection, analysis, interpretation, and visualization.

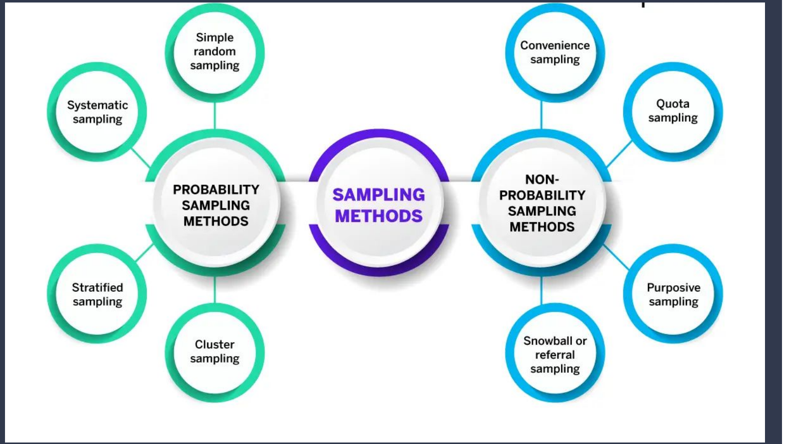
Statistics Terminology

Population: Population is the collection of individuals, objects, and events whose properties are to be analyzed.

Sample: A subset of the population is called a sample. A well-chosen sample contains most of the information about a particular population.

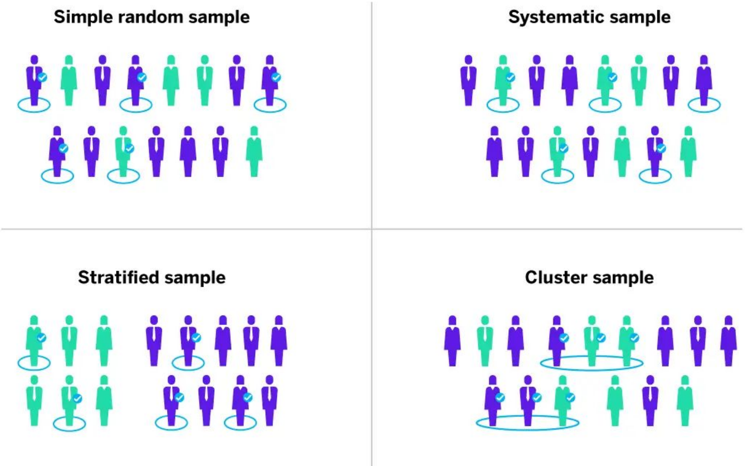


Sampling Techniques



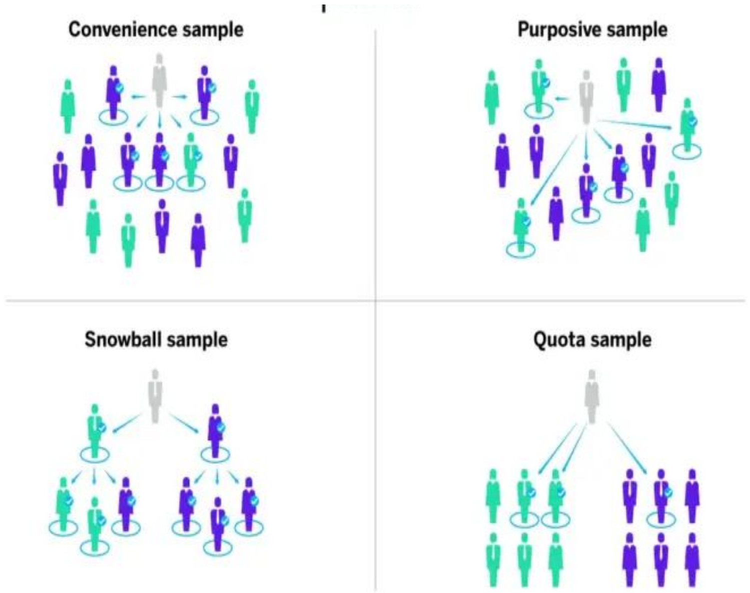
Probability Sampling:

- **Simple Random Sampling**
    - Each Member of the population has equal change being selected in the sample
- **Systematic Sampling**
    - In systematic sampling every nth record is chosen from the population.
- **Stratified Sampling**
    - Random sampling is used for select sufficient number subject from stratum.
- **Cluster Sampling**



Non probability Sampling

- **Convenience sampling**
  - elements in a sample are selected on the basis of their accessibility and availability.
- **Quota sampling**
  - A quota might include a certain number of males and a certain number of females. Alternatively, you might want your samples to be at a specific income level or in certain age brackets or ethnic groups.
- **Purposive sampling (Judgement)**
  - In judgement sampling, participants for the sample are chosen consciously by researchers based on their knowledge and understanding of the research question at hand or their goals.
- **Snowball sampling**
  - With this approach, people recruited to be part of a sample are asked to invite those they know to take part, who are then asked to invite their friends and family and so on.



Types of Statistics

1. Descriptive Statistics
2. Inferential Statistics

Descriptive Statistics

Descriptive statistics uses data that provides a description of the population either through numerical calculated graphs or tables.

I. Measure of Central tendency
   - Mean
   - Median
   - Mode

II. Measure of Variability (Spread)
   - Range
   - Interquartile Range
   - Variation
   - Standard Deviation

I. Measure of Central tendency

- **Mean**

mean is only the average of all numbers in a particular numeric variable.

Mean = ∑x/n

- **Median**

the median is a centre value after sorting all the numbers. The formula used to calculate the median of the data set is, suppose we are given ‘n’ terms in s data set,
If n is Even
Median = n/2
If n is Odd
Median = (n + 1)/2

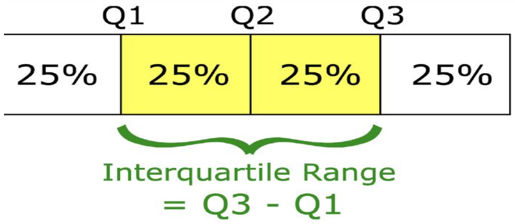
- **Mode**
Mode represents the most frequent observation in a numeric variable.

Measure of Variability (Spread)

Range
It is a given measure of how to spread apart values in a sample set or data set.

Range = Maximum value – Minimum value

Interquartile Range(IQR)
It is a measure of dispersion between upper(75th) and lower(25th) Quartiles. It is a very important term in statistics that is used in most calculations and data preprocessing like dealing with outliers.



Variance
It is used to measure the variability in the data from the mean.

$$\sigma^2 = \sum_{i=1}^{n} [(x_i - \bar{x})^2 / n]$$

- *n represents total data points*
- *$\bar{x}$ represents the mean of data points*
- *x_i represents individual data points*

Standard Deviation
It is the measure of the dispersion of a set of data from its mean.

$$\sigma = \sqrt{(1/n) \sum_{i=1}^{n} (x_i - \mu)^2}$$

Probability

Probability is the measure of how likely an event will occur

- ➢ Probability of all outcomes always sum to 1

Example:

- ➢ On a rolling a dice, you get 6 possible outcome.
- ➢ Each possibility only has one outcome, so each has a probability of ⅙.

Terminology in Probability

Random Experiment: An experiment or process for which outcome can not be predicted with certainty.

Sample Space: Entire possible set of outcome of a random experiment is the sample space of that experiment.

Event: One or more outcomes of an experiment it is the subset of sample space.

Types of Events

Disjoint Event: Do not have any common outcome

- The outcome of the ball can not be six and a wicket
- A man can not be dead and alive

Non Disjoint Event: can have common outcome

- The outcome of the ball can be a six or no ball

Types of Probability

- Marginal Probability
- Joint Probability
- Conditional Probability

- **Marginal Probability:**

Marginal probability is the probability of the occurrence of a single event.
Example: The probability of drawing a heart card in a deck is 13/52.

- **Joint Probability:**

It is the measure of two events happening at the same time **(p(A∩B)).**
Example: The probability that a card is a four and red = p(four and red) = 2/52=1/26. (There are two red fours in a deck of 52, the 4 of hearts and the 4 of diamonds).

- **Conditional Probability**

The conditional probability of an event B is the probability that the event will occur given the knowledge that an event A has already occurred. It is denoted by P(B|A).

➢ If A and B are dependent events, then the expression for the conditional probability is given by **P(B|A)=P(A and B)/P(A).**
➢ If A and B are independent, P(B|A)=P(B).

Regression

Regression is a supervised learning technique which helps in finding the correlation between variables and enables us to predict the continuous output variable based on the one or more predictor variables. It is mainly used for prediction, forecasting, time series modeling, and determining the causal-effect relationship between variables.

Basic Terminology:

Dependent Variable: The main factor in Regression analysis which we want to predict or understand is called the dependent variable. It is also called target variable.
Independent Variable: The factors which affect the dependent variables or which are used to predict the values of the dependent variables are called independent variable, also called as a predictor.

Types of Regression Techniques

- Linear Regression
- Multiple linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression
- Elastic Net Regression
- Decision Tree Regression
- Random Forest Regression
- Support Vector Regression

Simple Linear Regression

- It is a commonly used algorithm and can be imported from the Linear Regression class.
- A single input variable(the significant one) is used to predict one or more output variables, assuming that the input variable isn't correlated with each other.
- It is represented as : **y= m*x + c**
- Where y- dependent variable, x-independent, m-slope of the best fit line that could get accurate output and c -its intercept.
- Unless there is an exact line that relates the dependent and independent variables there might be a loss in output which is usually taken as the square of the difference between the predicted and actual output, ie the loss function or cost function.

When you use more than one independent variable to get output, it is termed Multiple linear regression. This kind of model assumes that there is a linear relationship between the given feature and output, which is its limitation.

Ridge and Lasso Regression

Ridge Regression-The L2 Norm

This is a kind of algorithm that is an extension of a linear regression that tries to minimize the loss, also uses multiple regression data. Its coefficients are not estimated by ordinary least squares (OLS), but by an estimator called ridge, which is biased and has lower variance than the OLS estimator thus we get shrinkage in coefficients. With this kind of model, we can reduce the model complexity as well.

$$j(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (h\theta^i - y^i)^2 + \lambda(slope)^2$$

Lasso Regression -The L1 Norm

It is The Least Absolute Shrinkage and Selection Operator. This penalizes the sum of absolute values of the coefficients to minimize the prediction error. It causes the regression coefficients for some of the variables to shrink to Zero. It can be constructed using the LASSO class. One of the advantages of the lasso is its simultaneous feature selection. This helps in minimizing the prediction loss.

$$j(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (h\theta^i - y^i)^2 + \lambda|slope|$$

Both lasso and ridge are regularisation method

Polynomial Regression

- This is an extension of linear regression and is used to model a non-linear relationship between the dependent variable and independent variables.
- Here as well syntax remains the same but now in the input variables we include some polynomial or higher degree terms of some already existing features as well.
- Linear regression was only able to fit a linear model to the data at hand but with polynomial features, we can easily fit some non-linear relationship between the target as well as input features.

The polynomial regression model of degree r is given by

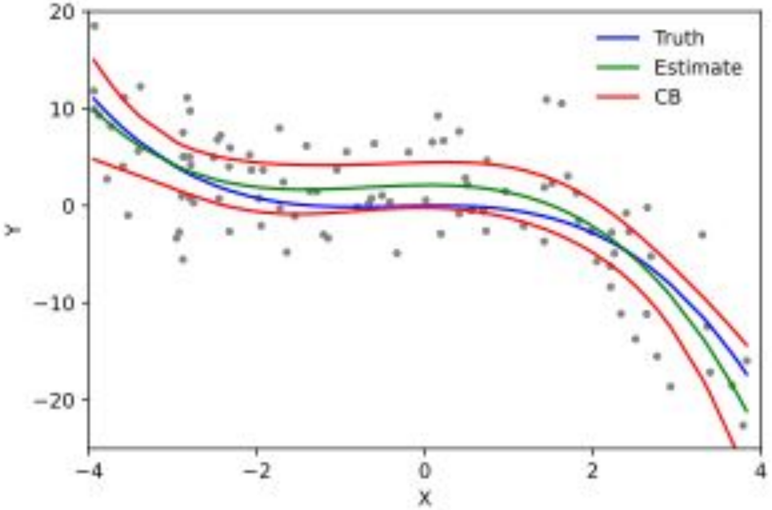
$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \cdots + \beta_r X_i^r + u_i.$$

(a) Interpret the coefficient β_1 in a linear regression ($r = 1$):

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

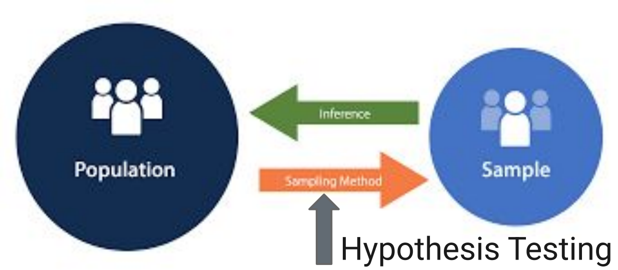
and in a quadratic regression ($r = 2$):

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + u_i.$$



Hypothesis Testing

- Hypothesis testing is a form of statistical inference that uses data from a sample to draw conclusions about a population parameter.
- First, a tentative assumption is made about the parameter. This assumption is called the null hypothesis and is denoted by H0.
- An alternative hypothesis (denoted Ha), which is the opposite of what is stated in the null hypothesis, is then defined.
- The hypothesis-testing procedure involves using sample data to determine whether or not H0 can be rejected.
- If H0 is rejected, the statistical conclusion is that the alternative hypothesis Ha is true.



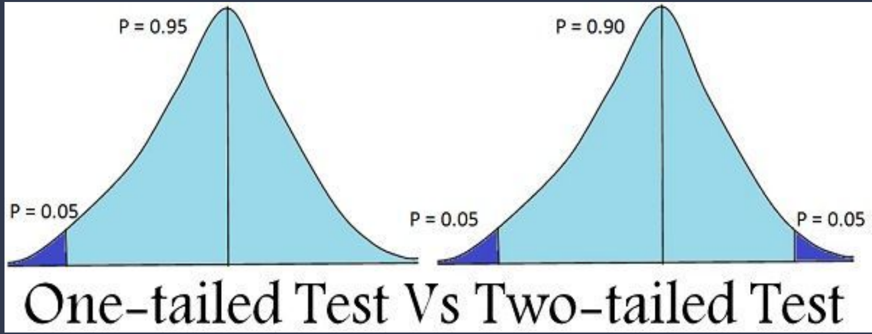
Hypothesis Testing

For example, assume that a radio station selects the music it plays based on the assumption that the average age of its listening audience is 30 years.

- To determine whether this assumption is valid, a hypothesis test could be conducted with the null hypothesis given as H_0: μ = 30
- The alternative hypothesis given as H_a: μ ≠ 30.
- Based on a sample of individuals from the listening audience, the sample mean age, $\bar{x}$, can be computed and used to determine whether there is sufficient statistical evidence to reject H_0.
- Conceptually, a value of the sample mean that is “close” to 30 is consistent with the null hypothesis, while a value of the sample mean that is “not close” to 30 provides support for the alternative hypothesis.

Hypothesis Testing and Confidence Intervals

confidence intervals and hypothesis tests are inferential techniques that depend on approximating the sample distribution. Data from a sample is used to estimate a population parameter using confidence intervals. Data from a sample is used in hypothesis testing to examine a given hypothesis. We must have a postulated parameter to conduct hypothesis testing.



Types of Hypothesis Testing

Z Test

It usually checks to see if two means are the same (the null hypothesis). Only when the population standard deviation is known and the sample size is 30 data points or more, can a z-test be applied.

To calculate the z-score, we would use the following formula: **$z = (\bar{x} - \mu 0) / (\sigma / \sqrt{n})$**

T Test

A statistical test called a t-test is employed to compare the means of two groups. Only when the population standard deviation is known and the sample size is less than 30, can a t-test be applied.

To calculate the t-score, we would use the following formula: **$z = (\bar{x} - \mu 0) / (s / \sqrt{n})$**

Chi-Square

You utilize a Chi-square test for hypothesis testing concerning whether your data is as predicted. To determine if the expected and observed results are well-fitted, the Chi-square test analyzes the differences between categorical variables from a random sample.

The chi-square formula is: **$\chi 2 = \sum(Oi - Ei)2/Ei$**, where Oi = observed value (actual value) and Ei = expected value.

Data Preprocessing

- Data preprocessing in Machine Learning refers to the technique of preparing (cleaning and organizing) the raw data to make it suitable for a building and training Machine Learning models.
- In simple words, data preprocessing in Machine Learning is a data mining technique that transforms raw data into an understandable and readable format.

Different Steps in Data Preprocessing :

1. Data Cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction

Steps in Data Preprocessing

Data Cleaning

Data Cleaning is particularly done as part of data preprocessing to clean the data by filling missing values, smoothing the noisy data, resolving the inconsistency, and removing outliers.

1. Missing values

Here are a few ways to solve this issue:

- Ignore those tuples
- Fill in the missing values

2. Noisy Data

It involves removing a random error or variance in a measured variable. It can be done with the help of the following techniques:

- Binning
- Regression
- Clustering

3. Removing outliers

Clustering techniques group together similar data points. The tuples that lie outside the cluster are outliers/inconsistent data.

Steps in Data Preprocessing continue..

Data Integration
Data Integration is one of the data preprocessing steps that are used to merge the data present in multiple sources into a single larger data store like a data warehouse.

Data Transformation
ML programmers must pay close attention to data transformation when it comes to data preprocessing steps. This process entails putting the data in a format that will allow for analysis. Normalization, standardization, and discretisation are common data transformation procedures. While standardization transforms data to have a zero mean and unit variance, normalization scales data to a common range. Continuous data is discretized into discrete categories using this technique.

Data Reduction
Data reduction is the process of lowering the dataset’s size while maintaining crucial information. Through the use of feature selection and feature extraction algorithms, data reduction can be accomplished.

Feature Scaling

- Feature scaling is a data preprocessing technique used to transform the values of features or variables in a dataset to a similar scale.
- Feature scaling becomes necessary when dealing with datasets containing features that have different ranges, units of measurement, or orders of magnitude.
- There are several common techniques for feature scaling, including standardization, normalization, and min-max scaling.
- These methods adjust the feature values while preserving their relative relationships and distributions.

Normalization:

- Normalization, a vital aspect of Feature Scaling, is a data preprocessing technique employed to standardize the values of features in a dataset, bringing them to a common scale.

- **Min Max Scaling**

Normalization is a scaling technique in which values are shifted and rescaled so that they end up ranging between 0 and 1. It is also known as Min-Max scaling.

Here's the formula:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

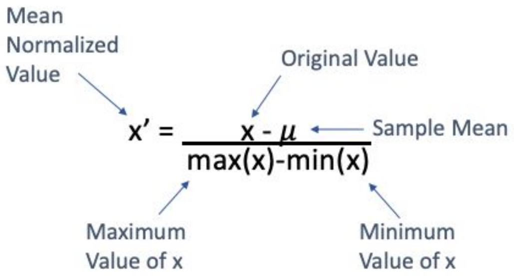
Normalization equation
Here, Xmax and Xmin are the maximum and the minimum values of the feature, respectively.

- When the value of X is the minimum value in the column, the numerator will be 0, and hence X' is 0
- On the other hand, when the value of X is the maximum value in the column, the numerator is equal to the denominator, and thus the value of X' is 1
- If the value of X is between the minimum and the maximum value, then the value of X' is between 0 and 1

Normalization continue..

- **Mean Normalization**

Mean normalization is a way to implement feature scaling. It involves calculating and subtracting the mean for each feature. It is also common practice to divide this value by the range or the standard deviation.

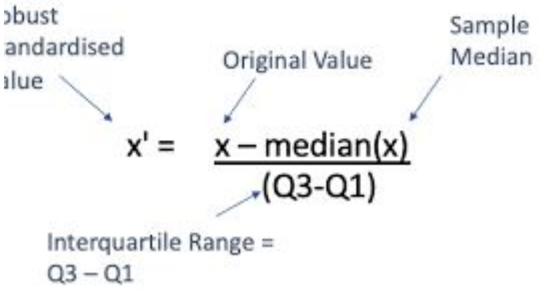


- **Max Absolute Scaling**

Maximum absolute scaling (Max Abs Scaling) is a technique in machine learning that normalizes numerical features. It scales data based on the maximum absolute value of each feature. Formula is: ***x/max(x)***

- **Robust Scaling**

Robust scaling is a machine learning algorithm that scales features that are resistant to outliers. It uses the median and interquartile range (IQR) instead of the mean and standard deviation. The scaled values have their median and IQR set to 0 and 1, respectively.



Data Visualization

- Data visualization is a crucial aspect of machine learning that enables analysts to understand and make sense of data patterns, relationships, and trends.
- Data visualization is an essential step in data preparation and analysis as it helps to identify outliers, trends, and patterns in the data that may be missed by other forms of analysis.

Types of Data Visualization

Line Charts: In a line chart, each data point is represented by a point on the graph, and these points are connected by a line.

Scatter Plots: A quick and efficient method of displaying the relationship between two variables is to use scatter plots.

Bar Charts: Bar charts are a common way of displaying categorical data. In a bar chart, each category is represented by a bar, with the height of the bar indicating the frequency or proportion of that category in the data.

Box Plots: Box plots are a graphical representation of the distribution of a set of data. In a box plot, the median is shown by a line inside the box, while the center box depicts the range of the data.

Histograms: This graph plots a distribution of numbers using a bar chart (with no spaces between the bars), representing the quantity of data that falls within a particular range.

Convex Optimization

- Convex optimization is a branch of optimization that works on minimizing a convex objective function subject to convex constraints.
- Convex optimization solution is crucial due to the many useful qualities that make it straightforward to solve and study.
- For instance, in the case of convex optimization problems, the optimal solution is guaranteed to exist in the form of a global minima.

Convex Set

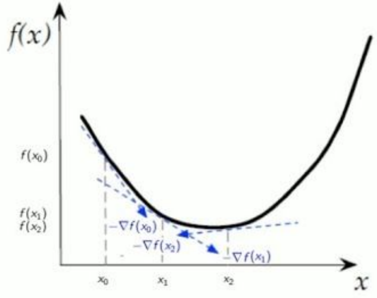
- A convex set is a collection of points in which the line AB connecting any two points A, B in the set lies completely within the set.

Convex Functions

A convex function is a function whose graph is always curved upwards, which means that the line segment connecting any two points on the graph is always above or on the graph itself. In other terms, a function f(x) is convex if and only if:

$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda) f(y)$

for any x and y in f's domain and any λ in the interval [0,1]. The convexity requirement is the defining feature of convex functions because of this inequality.



Topics for Self Study

- Logistic Regression
- Decision Tree
- Random Forest
- Bagging and Boosting
- K-mean Clustering

Online References:

- GeeksforGeeks
- Javatpoint
- Analytics Vidhya
- SimpliLearn

Youtube Channels :

- Krish Naik : https://youtube.com/playlist?list=PLZoTAELRMXVPBTrWtJkn3wWQxZkmTXGwe&si=xfov3UOzxtw02oLR
- CampusX : https://youtube.com/playlist?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH&si=hxU7n8MT03FjZ1ja